Exercise 1 — Concepts

Optimization-Based Control Methods · Linear Systems and Linear Controllers

Course: Optimization-Based Control Methods (TU Chemnitz, Automatic Control Group, Dr. Adrian Redder) **Topic:** Linear systems and linear controllers **Covers:** everything needed for Task 0 and Task 1 (Tasks 2, 3 and the numerical experiment will be added later)

This document explains every idea from the beginning, assuming no previous control-systems course. Every formula is derived, and no steps are skipped. The worked solutions are in `02_Solutions.md`.

Notation follows the lecture script: W_c is the controllability (reachability) matrix and W_o is the observability matrix.

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1. Linear algebra toolkit

Control theory for linear systems is mostly linear algebra. This section collects every tool used later.

1.1 Vectors and matrices

A **vector** $x \in \mathbb{R}^n$ is a column of n real numbers:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

A **matrix** $A \in \mathbb{R}^{n \times m}$ has n rows and m columns. The entry in row i, column j is a_{ij}.

The **transpose** $A^\top$ swaps rows and columns: $(A^\top)_{ij} = a_{ji}$. A column vector transposed becomes a row vector.

1.2 Matrix–vector and matrix–matrix products

Rule: entry i of Ax is the dot product of row i of A with x:

$$(Ax)_i = \sum_{j=1}^{m} a_{ij}\, x_j.$$

Example.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 0\cdot 0 + 0\cdot 1 \\ 0\cdot 1 + 2\cdot 0 + 2\cdot 1 \\ 0\cdot 1 + 1\cdot 0 + 3\cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Equivalent view (very useful): Ax is a *weighted sum of the columns of* A, with the weights being the entries of x:

$$Ax = x_1\,(\text{column } 1) + x_2\,(\text{column } 2) + \cdots + x_m\,(\text{column } m).$$

Row vector times matrix: $c^\top A$ is a weighted sum of the *rows* of A. For example $[1, 1, 1]\, A$ is the sum of all rows of A.

Matrix–matrix product: $(AB)_{ij}$ is row i of A dotted with column j of B. In general $AB \neq BA$.

Matrix powers: $A^0 = I$, $A^1 = A$, $A^2 = AA$, $A^k = AA^{k-1}$. To compute A^2B, it is easiest to compute AB first and then $A(AB)$.

Identity matrix I: ones on the diagonal, zeros elsewhere, so $Ix = x$ for every x.

1.3 Linear combinations, span, linear independence

A **linear combination** of vectors $v_1, \ldots, v_k$ is $c_1v_1 + \cdots + c_kv_k$ with real numbers c_i.

The **span** of $v_1, \ldots, v_k$ is the set of *all* their linear combinations. In $\mathbb{R}^3$:

- the span of one nonzero vector is a **line** through the origin;
- the span of two non-parallel vectors is a **plane** through the origin;
- the span of three independent vectors is **all of** $\mathbb{R}^3$.

The vectors are **linearly independent** if the only way to get $c_1v_1 + \cdots + c_kv_k = 0$ is $c_1 = \cdots = c_k = 0$. Otherwise, one of them can be written as a combination of the others, and it adds nothing new to the span.

1.4 Rank

The **rank** of a matrix is the number of linearly independent columns. It always equals the number of linearly independent rows. For $M \in \mathbb{R}^{n\times m}$: $\text{rank}(M) \leq \min(n, m)$. The matrix has **full rank** when equality holds.

How to compute the rank by hand (row reduction / Gaussian elimination). Allowed operations, which never change the rank:

1. swap two rows;
2. multiply a row by a nonzero number;
3. add a multiple of one row to another row.

Use them to create zeros below the leading entries ("echelon form"). The rank is the number of nonzero rows left.

Example.

$$\begin{bmatrix}1 & 1 & 1\\ 0 & 2 & 10\\ 1 & 3 & 11\end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix}1 & 1 & 1\\ 0 & 2 & 10\\ 0 & 2 & 10\end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix}1 & 1 & 1\\ 0 & 2 & 10\\ 0 & 0 & 0\end{bmatrix}$$

There are two nonzero rows, so the rank is 2.

Another test for square matrices: an $n \times n$ matrix has rank n if and only if its determinant is nonzero (Section 1.7).

1.5 Column space (image) and null space (kernel)

For a matrix M:

- **Column space / image:** $\text{im}(M) = \{Mv : v \text{ any vector}\}$, the span of the columns. Its dimension is $\text{rank}(M)$.
- **Null space / kernel:** $\ker(M) = \{v : Mv = 0\}$, all vectors that M sends to zero.

Rank–nullity theorem. For M with m columns:

$$\text{rank}(M) + \dim \ker(M) = m.$$

How to find a kernel: solve $Mv = 0$ by row reduction, express the "pivot" variables in terms of the free variables, and read off a basis.

Example. $\ker \begin{bmatrix}1 & 1 & 1\\ 1 & 3 & 5\end{bmatrix}$:

$$\begin{aligned} v_1 + v_2 + v_3 &= 0 \\ v_1 + 3v_2 + 5v_3 &= 0 \end{aligned} \quad \xrightarrow{\text{subtract}} \quad 2v_2 + 4v_3 = 0 \;\Rightarrow\; v_2 = -2v_3.$$

Then $v_1 = -v_2 - v_3 = 2v_3 - v_3 = v_3$. With $v_3 = t$: $v = t\,[1, -2, 1]^\top$. The kernel is the line spanned by $[1, -2, 1]^\top$.

How to test if a vector w lies in $\text{im}(M)$: $w \in \text{im}(M)$ if and only if $\text{rank}([M \;\; w]) = \text{rank}(M)$. Appending w must not add a new independent direction.

1.6 Orthogonality, orthogonal complement, cross product

Two vectors are **orthogonal** if their dot product is zero: $v^\top w = \sum_i v_i w_i = 0$.

The **orthogonal complement** $S^\perp$ of a subspace S is the set of all vectors orthogonal to every vector in S. In $\mathbb{R}^3$, the complement of a line is a plane and vice versa.

A key fact: **the row space of M is the orthogonal complement of** $\ker(M)$, because $Mv = 0$ says exactly that v is orthogonal to every row of M.

Describing a plane through the origin in $\mathbb{R}^3$ by one equation. If the plane is spanned by p and q, a **normal vector** is the cross product

$$p \times q = \begin{bmatrix} p_2 q_3 - p_3 q_2 \\ p_3 q_1 - p_1 q_3 \\ p_1 q_2 - p_2 q_1 \end{bmatrix},$$

and the plane is $\{x : (p \times q)^\top x = 0\}$.

1.7 Determinants

The determinant is a number $\det(M)$ attached to a square matrix. The facts we need:

- $\det(M) \neq 0 \iff M$ is invertible $\iff \operatorname{rank}(M) = n \iff \ker(M) = \{0\}$.
- $\det(M) = 0 \iff$ there is a $v \neq 0$ with $Mv = 0$.

2×2**:**

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

3×3 **— cofactor (Laplace) expansion along a row.** Each position has a sign from the checkerboard

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}, \qquad \text{sign of position } (i, j) = (-1)^{i+j}.$$

Pick any row. For each entry of that row: take the entry, multiply by its checkerboard sign, multiply by the 2×2 determinant that remains after deleting that entry's row and column (the **minor**). Add the three results. Along the first row:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = +a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \det \begin{bmatrix} d & e \\ g & h \end{bmatrix}.$$

Rules of thumb.

- Any row or column gives the same answer, so **expand along the one with the most zeros**: each zero kills a whole term.
- **Triangular matrix** (all zeros above *or* all zeros below the diagonal): the determinant is the product of the diagonal entries.
- $\det(AB) = \det(A) \det(B)$, $\det(A^\top) = \det(A)$.

1.8 Matrix inverse

M^{-1} is the matrix with $MM^{-1} = M^{-1}M = I$. It exists exactly when $\det M \neq 0$. Then $Mx = c$ has the unique solution $x = M^{-1}c$.

2×2 **formula:**

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

For a scalar, $m^{-1} = 1/m$, which exists when $m \neq 0$.

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2. Eigenvalues and eigenvectors

2.1 Definition and meaning

A nonzero vector v is an **eigenvector** of a square matrix A, with **eigenvalue** λ, if

$$Av = \lambda v.$$

Most vectors change direction when multiplied by A. Eigenvectors don't: A only stretches them by the factor λ. If λ is negative, the vector also flips. If λ is complex, the state rotates.

Why control engineers care. If $x_0 = v$ and $x_{k+1} = Ax_k$, then

$$x_1 = Av = \lambda v, \quad x_2 = A(\lambda v) = \lambda^2 v, \quad \dots, \quad x_k = \lambda^k v.$$

So along an eigenvector, the state is simply multiplied by λ every step. Each eigenvalue is a **mode** of the system. The lecture also calls the eigenvalues of A the **poles** of the system.

2.2 Deriving the characteristic equation

Start from $Av = \lambda v$ and rewrite it:

$$Av = \lambda v \iff Av = \lambda I v \iff \lambda I v - Av = 0 \iff (\lambda I - A)\, v = 0.$$

(We wrote $\lambda v = \lambda I v$ so that we can factor out v. "$\lambda - A$" alone would not make sense, since a number minus a matrix is undefined.)

We need a **nonzero** v solving $(\lambda I - A)v = 0$. By Section 1.7, such a v exists exactly when $\lambda I - A$ is singular:

$$\boxed{\det(\lambda I - A) = 0} \qquad \text{(characteristic equation)}$$

$p(\lambda) = \det(\lambda I - A)$ is a polynomial of degree n in λ, called the **characteristic polynomial**. Its n roots (counted with multiplicity, possibly complex) are the eigenvalues.

2.3 Procedure

1. **Form $\lambda I - A$:** put λ on the diagonal and flip the sign of *every* entry of A. The diagonal becomes $\lambda - a_{ii}$ and an off-diagonal entry a_{ij} becomes $-a_{ij}$.
2. **Take the determinant** (expand along the row or column with the most zeros).
3. **Find the roots** of the polynomial (factor it, or use the quadratic formula).
4. **Check** with trace and determinant (Section 2.5).
5. *(If needed)* **eigenvectors:** for each λ, solve $(\lambda I - A)v = 0$, which is a kernel computation as in Section 1.5.

2.4 The 2×2 shortcut

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - a & -b \\ -c & \lambda - d \end{bmatrix} = (\lambda - a)(\lambda - d) - (-b)(-c) = \lambda^2 - (a+d)\lambda + (ad - bc).$$

Since $a + d = \mathrm{tr}(A)$ (the trace, i.e. the sum of the diagonal entries) and $ad - bc = \det(A)$:

$$\boxed{\lambda^2 - \mathrm{tr}(A)\,\lambda + \det(A) = 0} \qquad \Rightarrow \qquad \lambda_{1,2} = \frac{\mathrm{tr}(A) \pm \sqrt{\mathrm{tr}(A)^2 - 4\det(A)}}{2}.$$

2.5 Checks that always hold (any size)

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \mathrm{tr}(A), \qquad \lambda_1 \cdot \lambda_2 \cdots \lambda_n = \det(A).$$

Use them every time to catch arithmetic mistakes.

2.6 Special structures (shortcuts)

- **Diagonal or triangular matrix:** the eigenvalues are the diagonal entries. *Why:* $\lambda I - A$ is also triangular, so its determinant is the product of its diagonal: $(\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn})$. The entries off the diagonal are always multiplied by a zero during expansion.
- **Block structure:** if row i and column i are zero except for the diagonal entry a_{ii}, then a_{ii} is an eigenvalue. The remaining eigenvalues are those of the smaller matrix left after deleting row i and column i. More generally, for block-triangular matrices, the eigenvalues are the eigenvalues of the diagonal blocks.
- **Complex eigenvalues:** if $\mathrm{tr}(A)^2 - 4\det(A) < 0$, the roots are $\lambda = \alpha \pm i\beta$. Their magnitude is $|\lambda| = \sqrt{\alpha^2 + \beta^2}$. Complex eigenvalues mean rotation or oscillation.
- **Repeated eigenvalues:** a root that appears more than once, e.g. $(\lambda - 3)^2$. Its **algebraic multiplicity** is how often it is a root. Its **geometric multiplicity** is $\dim \ker(\lambda I - A)$, the number of independent eigenvectors.

2.7 Worked example

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix}, \qquad \lambda I - A = \begin{bmatrix} \lambda - 1 & 0 & 0 \\ 0 & \lambda - 2 & -2 \\ 0 & -1 & \lambda - 3 \end{bmatrix}.$$

Expand along row 1, which has two zeros:

$$\det(\lambda I - A) = (\lambda - 1)\det \begin{bmatrix} \lambda - 2 & -2 \\ -1 & \lambda - 3 \end{bmatrix} = (\lambda - 1)\big[(\lambda - 2)(\lambda - 3) - (-2)(-1)\big].$$

Expand the bracket: $(\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$, and $(-2)(-1) = 2$, so the bracket is $\lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4)$. Therefore

$$\det(\lambda I - A) = (\lambda - 1)^2(\lambda - 4) \quad \Rightarrow \quad \lambda = 1,\, 1,\, 4.$$

Check: $1 + 1 + 4 = 6 = 1 + 2 + 3 = \mathrm{tr}(A)$ ✓ and $1 \cdot 1 \cdot 4 = 4 = \det(A)$ ✓.

2.8 Cayley–Hamilton theorem

Every square matrix satisfies its own characteristic polynomial: if $p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$, then

$$p(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = 0.$$

Consequence: $A^n = -c_{n-1}A^{n-1} - \cdots - c_0 I$, so A^n, and therefore every higher power, is a linear combination of $I, A, \ldots, A^{n-1}$. We use this in Section 5 to show that n steps are always enough.

Example: for the A above, $p(\lambda) = (\lambda-1)^2(\lambda-4) = \lambda^3 - 6\lambda^2 + 9\lambda - 4$, so $A^3 = 6A^2 - 9A + 4I$.

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3. Discrete-time LTI systems

3.1 The model

$$\boxed{\begin{aligned} x_{k+1} &= Ax_k + Bu_k \\ y_k &= Cx_k \ (+ Du_k) \end{aligned}}$$

Symbol	Name	Size	Meaning
k	time index	$k = 0, 1, 2, \ldots$	sample number; real time is $t = kT_s$ with sampling time T_s
x_k	state	n	everything you need to know about the system at time k
u_k	input	m	what the controller commands (voltage, force, ...)
y_k	output	p	what the sensors measure
A	system matrix	$n \times n$	how the state evolves by itself
B	input matrix	$n \times m$	how inputs enter the state
C	output matrix	$p \times n$	which combination of states the sensors see
D	feedthrough	$p \times m$	direct input-to-output path (often 0)

- **Linear:** the right-hand sides are linear in x_k and u_k. So superposition holds: the response to a sum of inputs or initial states is the sum of the individual responses.
- **Time-invariant (LTI):** A, B, C, D are constant and do not depend on k.
- **Single-input (SISO)** if $m = 1$, **multiple-input (MIMO)** if $m > 1$. The number of inputs is the number of columns of B.

3.2 Solution formula (derived by unrolling)

Apply the update equation repeatedly:

$$\begin{aligned} x_1 &= Ax_0 + Bu_0 \\ x_2 &= Ax_1 + Bu_1 = A(Ax_0 + Bu_0) + Bu_1 = A^2x_0 + ABu_0 + Bu_1 \\ x_3 &= Ax_2 + Bu_2 = A^3x_0 + A^2Bu_0 + ABu_1 + Bu_2. \end{aligned}$$

The pattern: input u_j is applied at time j and then gets multiplied by A once for every remaining step until time k, which is $k - 1 - j$ times. Hence

$$x_k = \underbrace{A^k x_0}_{\text{free response}} + \underbrace{\sum_{j=0}^{k-1} A^{k-1-j} B u_j}_{\text{forced response}}$$

(Formally: true for $k = 0$; if true for k, then $x_{k+1} = Ax_k + Bu_k = A^{k+1}x_0 + \sum_{j=0}^{k-1} A^{k-j}Bu_j + Bu_k$, which is the formula for $k+1$.)

3.3 Modes: why eigenvalues decide everything

Suppose A has n independent eigenvectors $v_1, \dots, v_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$ (this holds, for example, when all eigenvalues are distinct). Write the initial state as a combination $x_0 = c_1 v_1 + \cdots + c_n v_n$. With $u = 0$, using $A^k v_i = \lambda_i^k v_i$:

$$x_k = A^k x_0 = c_1 \lambda_1^k v_1 + \cdots + c_n \lambda_n^k v_n.$$

Each term behaves like λ_i^k:

λ	λ^k as $k \to \infty$
$\lvert\lambda\rvert < 1$	$\to 0$ (decays)
$\lvert\lambda\rvert = 1$	stays the same size ($\lambda = 1$: constant; $\lambda = -1$: flips sign; complex: rotates)
$\lvert\lambda\rvert > 1$	grows without bound
$\lambda = 0$	is exactly 0 after one step

> In discrete time, the dividing line is the **unit circle** $\lvert\lambda\rvert = 1$. In continuous time it would instead be the imaginary axis, Re $\lambda = 0$. Don't mix the two.

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4. Stability

4.1 Definitions (for $x_{k+1} = Mx_k$)

- **Asymptotically stable:** $x_k \to 0$ for every x_0. $\iff$ **all eigenvalues satisfy** $\lvert\lambda\rvert < 1$ (strictly inside the unit circle). $\iff$ $M^k \to 0$ as $k \to \infty$.
- **Lyapunov (marginally) stable:** x_k stays bounded for every x_0, but need not go to 0. This holds when all $\lvert\lambda\rvert \leq 1$ and every eigenvalue on the unit circle has enough independent eigenvectors (geometric multiplicity = algebraic multiplicity). The lecture phrases this as: eigenvalues inside the unit circle, with at least one of absolute value 1.
- **Unstable:** some x_0 gives an unbounded x_k. This happens, for example, if any $\lvert\lambda\rvert > 1$.

In the lecture, "the matrix M is asymptotically stable" is shorthand for "the trajectories of $x_{k+1} = Mx_k$ are asymptotically stable".

4.2 The geometric (Neumann) series and Proposition 1.1

Lemma. If all eigenvalues of M satisfy $\lvert\lambda\rvert < 1$, then $I - M$ is invertible and

$$\sum_{j=0}^{\infty} M^j = (I - M)^{-1}.$$

Proof, step by step.

1. **Invertibility.** If $I-M$ were singular, there would be $v \neq 0$ with $(I-M)v = 0$, i.e. $Mv = v$. Then 1 would be an eigenvalue of M, which contradicts $|\lambda| < 1$. So $I - M$ is invertible.
2. **Telescoping sum.** Multiply out:

$$\Big(\sum_{j=0}^{k} M^j\Big)(I - M) = (I + M + \cdots + M^k) - (M + M^2 + \cdots + M^{k+1}) = I - M^{k+1}.$$

   Every middle term appears once with $+$ and once with $-$, so they cancel.
3. **Take the limit.** Since M is asymptotically stable, $M^{k+1} \to 0$, so $(\sum_{j=0}^{\infty} M^j)(I - M) = I$.
4. Multiply both sides on the right by $(I - M)^{-1}$. ■

This is the matrix version of $1 + q + q^2 + \cdots = \frac{1}{1-q}$ for $|q| < 1$.

Proposition 1.1 (lecture). If M has all eigenvalues strictly inside the unit circle and the input is constant, $u_k = u$, then $x_{k+1} = Mx_k + Bu$ converges, for any x_0, to

$$x_\infty = (I - M)^{-1} Bu.$$

Proof. By the solution formula with constant input, $x_k = M^k x_0 + \sum_{j=0}^{k-1} M^{k-1-j} Bu = M^k x_0 + (\sum_{i=0}^{k-1} M^i) Bu$, after re-indexing with $i = k-1-j$. As $k \to \infty$: $M^k x_0 \to 0$ and the sum tends to $(I - M)^{-1}$. ■

4.3 Equilibria (fixed points)

A state $\bar{x}$ is an **equilibrium** of $x_{k+1} = f(x_k)$ if $\bar{x} = f(\bar{x})$: once there, you stay there.

How to find it: set $x_{k+1} = x_k = \bar{x}$ and solve. For $x_{k+1} = Mx_k + c$ with constant c:

$$\bar{x} = M\bar{x} + c \iff (I - M)\bar{x} = c \iff \bar{x} = (I - M)^{-1} c \quad (\text{if } I - M \text{ is invertible}).$$

Important: *if* a trajectory converges to some limit x_∞, then x_∞ must be an equilibrium. Take the limit on both sides of $x_{k+1} = Mx_k + c$ to get $x_\infty = Mx_\infty + c$.

4.4 The "shift to the equilibrium" trick

To prove that $x_{k+1} = Mx_k + c$ actually converges to $\bar{x}$, define the deviation $d_k := x_k - \bar{x}$. Then

$$d_{k+1} = x_{k+1} - \bar{x} = (Mx_k + c) - (M\bar{x} + c) = M(x_k - \bar{x}) = Md_k.$$

The constant c cancels. So $d_k = M^k d_0 \to 0$ when M is asymptotically stable, which means $x_k \to \bar{x}$. This trick is used in Tasks 1, 2 and 3.

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5. Reachability, controllability, stabilizability

Question: **can the input u steer the state where we want?**

5.1 Deriving the controllability matrix W_c

From the solution formula with $k = n$:

$$x_n = A^n x_0 + A^{n-1}Bu_0 + A^{n-2}Bu_1 + \cdots + ABu_{n-2} + Bu_{n-1}.$$

Collect the coefficient matrices side by side and the inputs into one long vector:

$$x_n - A^n x_0 = \underbrace{\begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}}_{W_c \in \mathbb{R}^{n \times nm}} \underbrace{\begin{bmatrix} u_{n-1} \\ u_{n-2} \\ \vdots \\ u_0 \end{bmatrix}}_{\vec{u}}.$$

Note the **reversed order**: the *last* input u_{n-1} multiplies B, and the *first* input u_0 multiplies $A^{n-1}B$, because it has had the most steps to be multiplied by A.

This is a linear system of equations $W_c\vec{u} = b$ with $b = x_n - A^n x_0$. It has a solution for **every** right-hand side b exactly when the columns of W_c span all of $\mathbb{R}^n$, i.e. when $\text{rank}(W_c) = n$.

5.2 Definitions (as in the lecture)

> **Reachability (fully state controllable), Def. 1.1:** the system is reachable $\iff$ $\text{rank}(W_c) = n$. Then any x_0 can be steered to any x^* in finite time.
>
> **Controllability, Def. 1.2:** the origin $x = 0$ can be reached from any initial state x_0 in finite time.
>
> **Stabilizability, Def. 1.7:** there exists K such that $A - BK$ is asymptotically stable.

Reachable $\Rightarrow$ Controllable $\Rightarrow$ Stabilizable, (the reverse implications are false in general).

Controllable but not reachable is possible when the "uncontrollable" part dies out on its own in finite time, i.e. it has eigenvalue 0. Example: A nilpotent ($A^k = 0$ for some k) and $B = 0$.

5.3 Why n steps are always enough (Cayley–Hamilton)

From Section 2.8, $A^n B$ is a combination of $B, AB, \ldots, A^{n-1}B$. The same holds for $A^{n+1}B$, etc. So **adding more steps never adds new directions**: the reachable set after n steps is the reachable set after any number of steps.

This also explains a pattern you'll see in practice: once $A^j B$ depends on the previous columns, all later columns do too.

5.4 The reachable subspace

Starting from $x_0 = 0$, the states reachable in N steps are

$$x_N = \sum_{j=0}^{N-1} A^{N-1-j} B u_j \in \text{span}\{B, AB, \dots, A^{N-1}B\}.$$

Reachable subspace $= \text{im}(W_c)$, the column space of W_c. Its dimension is $\text{rank}(W_c)$.

- A target x_f is reachable from $0 \iff x_f \in \text{im}(W_c) \iff \text{rank}[W_c \;\; x_f] = \text{rank}(W_c)$.
- **Minimum number of steps N:** the smallest N with $x_f \in \text{span}\{B, AB, \dots, A^{N-1}B\}$. Try $N = 1, 2, \dots$ in turn.
- **Input sequence:** solve $x_f = A^{N-1}Bu_0 + \cdots + ABu_{N-2} + Bu_{N-1}$ for $u_0, \dots, u_{N-1}$.

5.5 Conserved quantities: an intuitive picture of an unreachable direction

If $\text{rank}(W_c) < n$, there is a nonzero vector w orthogonal to the reachable subspace, $w^\top W_c = 0$. In particular $w^\top B = 0$. Suppose additionally $w^\top A = \lambda\, w^\top$ (w is a **left eigenvector**). Then the scalar $z_k := w^\top x_k$ evolves as

$$z_{k+1} = w^\top A x_k + w^\top B u_k = \lambda\, w^\top x_k + 0 = \lambda z_k.$$

The input has no influence on z_k at all. If $\lambda = 1$, then z_k is constant forever, a **conserved quantity**. That is exactly what makes a direction uncontrollable.

5.6 PBH (Popov–Belevitch–Hautus) eigenvalue test

$$\text{Eigenvalue } \lambda \text{ of } A \text{ is uncontrollable} \iff \text{rank}\,[\lambda I - A \quad B] < n.$$

Why: the rank drops exactly when some $w \neq 0$ satisfies $w^\top(\lambda I - A) = 0$ and $w^\top B = 0$. That is precisely the situation of Section 5.5: a left eigenvector that the input cannot touch.

You only need to test the eigenvalues of A. For any other λ, $\lambda I - A$ alone already has rank n.

Reading the PBH result:

Uncontrollable eigenvalues	Conclusion
none	reachable
only $\lambda = 0$	controllable (that part vanishes by itself in finite time), not reachable
only $\lvert\lambda\rvert < 1$	stabilizable (decays by itself asymptotically), but not controllable if $\lambda \neq 0$
some $\lvert\lambda\rvert \geq 1$	not even stabilizable

5.7 Pole placement (preview)

If the system is reachable, K can be chosen so that $A - BK$ has **any** desired eigenvalues. If it is only stabilizable, the uncontrollable eigenvalues stay where they are whatever K is, but the rest can be moved inside the unit circle.

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6. Observability and detectability

Question: **can the internal state be determined from the measured outputs?**

6.1 Deriving the observability matrix W_o

Without loss of generality set $u_k = 0$. The known input contribution can always be subtracted. Then $x_k = A^k x_0$ and

$$\begin{aligned} y_0 &= Cx_0, \\ y_1 &= Cx_1 = CAx_0, \\ &\vdots \\ y_{n-1} &= CA^{n-1}x_0. \end{aligned} \qquad \Longrightarrow \qquad \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}}_{W_o \in \mathbb{R}^{np \times n}} x_0.$$

Recovering x_0 means solving $W_o x_0 = \vec{y}$. The solution is unique exactly when $\ker(W_o) = \{0\}$, i.e. $\text{rank}(W_o) = n$. As before, Cayley–Hamilton says that more than n measurements add no new information.

6.2 Definitions

> **Unobservable state, Def. 1.3:** $z \neq 0$ is unobservable if starting at $x_0 = z$ with zero input produces $y_k = 0$ for all k.
>
> **Observability, Def. 1.4/1.5:** the system is observable $\iff \text{rank}(W_o) = n$.
>
> **Detectability, Def. 1.6:** every unobservable state decays to the origin by itself.

6.3 Unobservable and observable subspaces

- **Unobservable subspace** $= \ker(W_o)$: all x_0 with $W_o x_0 = 0$. These produce identically zero output, so they **cannot be distinguished from** $x_0 = 0$.
- **Observable subspace** $= \ker(W_o)^\perp$ = the row space of W_o.

Why this matters. Every x_0 splits uniquely as $x_0 = x_{\text{obs}} + x_{\text{unobs}}$, with x_{obs} in the observable subspace and x_{unobs} in $\ker(W_o)$. By linearity $\vec{y} = W_o x_{\text{obs}} + 0$, so the outputs determine only x_{obs}. If x_0 itself lies in the observable subspace ($x_{\text{unobs}} = 0$), it is the unique state in that subspace that explains the data. If it lies in the unobservable subspace, the output is zero and nothing can be learned.

Test: x_0 lies in the observable subspace $\iff$ x_0 is orthogonal to every basis vector of $\ker(W_o)$.

6.4 Duality and PBH test for observability

Observability is the mirror image of reachability:

$$(A, C) \text{ observable} \iff (A^\top, C^\top) \text{ reachable}, \qquad W_o(A, C) = W_c(A^\top, C^\top)^\top.$$

PBH test:

$$\text{Eigenvalue } \lambda \text{ is unobservable} \iff \text{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} < n \iff \exists\, v \neq 0 :\ Av = \lambda v,\ Cv = 0.$$

In words: an eigenvector that the sensor does not see. The system is **detectable** iff all unobservable eigenvalues satisfy $|\lambda| < 1$.

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7. State feedback, equilibria and tracking

7.1 State feedback and the closed loop

A **linear state-feedback controller** chooses the input from the measured state:

$$u_k = -Kx_k, \qquad K \in \mathbb{R}^{m\times n}.$$

Substituting it into the system:

$$x_{k+1} = Ax_k + B(-Kx_k) = (A - BK)\, x_k.$$

$A - BK$ is the **closed-loop system matrix**. Choosing K changes its eigenvalues. $x_k \to 0$ for all $x_0 \iff A - BK$ has all eigenvalues strictly inside the unit circle.

Open loop vs closed loop: an open-loop policy is a precomputed input sequence $u_0, u_1, \ldots$ that ignores the actual state. A closed-loop policy $u_k = \mu(x_k)$ reacts to the measured state, so it can correct disturbances and model errors.

7.2 Tracking control

Goal: make $x_k \to x_{\text{ref}}$ for a given constant reference. The natural first attempt is a **MIMO P-controller**:

$$u_k = -K(x_k - x_{\text{ref}}).$$

Closed loop:

$$x_{k+1} = Ax_k - BKx_k + BKx_{\text{ref}} = (A - BK)x_k + BK\, x_{\text{ref}}.$$

This has the form $x_{k+1} = Mx_k + c$ with $M = A - BK$ and constant $c = BKx_{\text{ref}}$. Sections 4.3 and 4.4 apply directly.

- **Tracking error:** $e_k = x_k - x_{\text{ref}}$.
- **Asymptotic (steady-state) tracking error:** $e_\infty = x_\infty - x_{\text{ref}}$.

7.3 Why a P-controller generally leaves a steady-state error

At steady state the state must be an equilibrium. To **hold** the system at some state $\bar{x}$, the input $\bar{u}$ must satisfy

$$\bar{x} = A\bar{x} + B\bar{u} \iff (I - A)\bar{x} = B\bar{u}.$$

So **holding x_{ref} requires an input $\bar{u}$ with $B\bar{u} = (I-A)x_{\text{ref}}$**, which is nonzero unless $Ax_{\text{ref}} = x_{\text{ref}}$. But at $x = x_{\text{ref}}$ the P-controller outputs $u = -K\cdot 0 = 0$. The controller can only produce a nonzero input when there is a nonzero error. The system therefore settles where the remaining error is just large enough to produce the input needed to hold it there.

Everyday analogy: a heater controlled by "power $\propto$ (target $-$ room temperature)" switches off exactly at the target. The room keeps losing heat, so it settles a little below the target.

The two standard fixes are the subject of Tasks 2 and 3:

1. **Reference prefilter F:** aim at a modified target Fx_{ref}, chosen so that the closed loop settles exactly at x_{ref}.

2. **Integral action:** add an integrator state $e^i_{k+1} = e^i_k + T_s e_k$ that keeps accumulating the error. It can only stop changing when $e_k = 0$. This is the discrete-time **MIMO PI controller**.

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8. Cheat sheet

Concept	Formula / test
Characteristic equation	$\det(\lambda I - A) = 0$
2×2 eigenvalues	$\lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$
Eigenvalue checks	$\sum \lambda_i = \mathrm{tr}\, A$, $\prod \lambda_i = \det A$
Solution	$x_k = A^k x_0 + \sum_{j=0}^{k-1} A^{k-1-j} B u_j$
Asymptotic stability (DT)	all $\lvert\lambda_i\rvert < 1$
Geometric series	$\sum_{j\geq 0} M^j = (I - M)^{-1}$ if M stable
Constant input limit	$x_\infty = (I - A)^{-1} B u$
Equilibrium of $x^+ = Mx + c$	$\bar{x} = (I - M)^{-1} c$
$I - M$ singular	$\iff$ 1 is an eigenvalue of M
Controllability matrix	$W_c = [B,\ AB,\ \ldots,\ A^{n-1}B]$
Reachable	$\mathrm{rank}\, W_c = n$
Reachable subspace	$\mathrm{im}(W_c)$
x_f reachable from 0	$\mathrm{rank}[W_c\ x_f] = \mathrm{rank}\, W_c$
PBH uncontrollable λ	$\mathrm{rank}[\lambda I - A,\ B] < n$
Controllable (DT)	all uncontrollable eigenvalues $= 0$
Stabilizable	all uncontrollable eigenvalues $\lvert\lambda\rvert < 1$
Observability matrix	$W_o = [C;\ CA;\ \ldots;\ CA^{n-1}]$
Observable	$\mathrm{rank}\, W_o = n$
Unobservable subspace	$\ker(W_o)$
Observable subspace	$\ker(W_o)^\perp$ = row space of W_o
PBH unobservable λ	$\mathrm{rank}[\lambda I - A;\ C] < n$
Detectable	all unobservable eigenvalues $\lvert\lambda\rvert < 1$
Closed loop (state feedback)	$x_{k+1} = (A - BK)x_k$
P-tracking closed loop	$x_{k+1} = (A - BK)x_k + BKx_{\mathrm{ref}}$
P-tracking limit	$x_\infty = (I - A + BK)^{-1} BK x_{\mathrm{ref}}$
P-tracking error	$x_\infty - x_{\mathrm{ref}} = (I - A + BK)^{-1}(A - I)x_{\mathrm{ref}}$